CEFOPERAZONE + SULBACTAM

Avoid alcohol (disulfiram-like)

Bleeding risk

CHEMICAL NAME Cefoperazone (piperazine cephalosporin) + sulbactam: (2S,5R)-3,3-dimethyl-7-oxo-4-thia-1-azabicyclo[3.2.0]heptane-2-carboxylic acid 4,4-dioxide

CLASS Third-generation cephalosporin + β-lactamase inhibitor

TRADE NAMES Sulperazon · Sulbacef · Magnex

THERAPEUTIC USES / INDICATIONS

- Intra-abdominal and biliary infections (high biliary excretion).
- Hospital-acquired pneumonia and respiratory tract infections.
- Complicated UTI, septicaemia, gynaecological and skin infections.
- *Acinetobacter* infections (sulbactam component).

ADMINISTRATION

- **IV** injection over 3–5 min or infusion over 15–60 min; **IM**.
- Incompatible with aminoglycosides in the same line.

DOSAGES

- **Adult (1:1):** 2–4 g/day divided every 12 h; severe up to 8 g/day.
- Max sulbactam 4 g/day.
- **Children:** 40–80 mg/kg/day divided every 6–12 h (max 160 mg/kg/day).
- Renal impairment: limit sulbactam dose.

PHARMACOKINETICS

- **Onset:** immediate IV.
- Cefoperazone: excreted mainly in bile; sulbactam: in urine.
- **t½:** about 1–2 h.

PHARMACODYNAMICS

- Cefoperazone inhibits cell-wall synthesis (incl. some *Pseudomonas* activity).
- Sulbactam inhibits β-lactamases and has intrinsic activity against *Acinetobacter*.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea, nausea.
- Rash; phlebitis.

ADVERSE EFFECTS (SERIOUS)

- Hypoprothrombinaemia and bleeding (N-methylthiotetrazole side chain → vitamin K antagonism).
- Disulfiram-like reaction with alcohol.
- Anaphylaxis; *C. difficile* colitis.
- Neutropenia.

CONTRAINDICATIONS & CAUTIONS

- Cephalosporin or β-lactam hypersensitivity.
- **Caution:** biliary obstruction, hepatic disease, malnutrition, bleeding disorders, anticoagulant use.

KEY INTERACTIONS

- **Alcohol (during and 72 h after):** flushing, vomiting, tachycardia.
- **Warfarin, antiplatelets:** ↑ bleeding.
- **Aminoglycosides:** incompatible in line.

NURSING MANAGEMENT

- Check allergy history.
- Monitor PT/INR in prolonged therapy; vitamin K may be prescribed.
- Watch for bleeding: gums, urine, stools, puncture sites.
- Avoid alcohol-containing medicines.
- Monitor LFTs, CBC and infection markers.

PATIENT EDUCATION

- No alcohol during treatment and for 3 days after.
- Report bleeding, bruising or black stools.
- Report severe diarrhoea.

KEY POINTS

- 3rd-gen cephalosporin + β-lactamase inhibitor.
- Bleeding risk (vitamin K) and alcohol reaction.
- Biliary excretion — good for biliary sepsis.
- Max sulbactam 4 g/day.